

## Major contributors to this report

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- **Dr Peter Asaro**, co-founder and vice chair, International Committee for Robot Arms Control

## Utopia or dystopia

EDI did not have a smooth experience on the battlefield. It chose a fictional attack scenario that, if implemented, would have caused a massive war to break out. By taking humans out of the loop of decision-making, this jet's logical reasoning could have been the cause of another war. But then, our entire history is littered with wars that were the direct results of human emotions and egos being involved. The question is, are egos and emotions more dangerous than logic?

99 per cent of the numbers used in a key calculation require only eight bits, so they were able to limit the resources devoted to it.

NVIDIA also announced a new chip called Tesla P100, which is designed for a technique called deep learning used by Google software AlphaGo. They also unveiled a special computer for researchers involved in deep learning. It comes with eight P100 chips, memory and storage. Models of this computer, known as DGX-1, will be sold for US\$ 129,000. It is being given to leading academic research groups, including ones at University of California, Berkeley, Stanford, New York University and MIT—all in USA.

## Solving the problem of skewed measurements

The two eyes humans possess help with depth perception. This is helpful in many cases. Similarly, when we talk about machines, we usually have single aperture systems, which raise the question of skewed measurements. Arvind Lakshmikummar, chief executive officer, Tonbo Imaging, says, "Aperture systems have their own set of limitations that can be overcome with multiple

aperture systems."

Efficient imaging systems employ sensors to the optimum. Their signals have to be safely transferred to the computing unit. Post imaging, signal transmission is another part to be looked at. "Intelligence guys often tap signals," says Parag Naik, co-founder and chief executive officer, Saankhya Labs.

Hence, signal security becomes an issue while transmitting the signals as well. Some security measures can be taken by making suitable changes in the system. However, even with the best of systems, "chances are still pretty high for collateral damage," says Lakshmikummar.

## How does somebody go about banning weapons

The open letter signed by Elon Musk, Stephen Hawking and Steve Wozniak, among others, made big news last year. It warned the world about the dangers of weaponised robots and a military AI arms race currently taking place between the world's military powers. The section on autonomous weapons reads, "If it is permissible or legal to use lethal autonomous weapons, how should these weapons be integrated into the existing command and control structure so that responsibility and liability remain associated with specific human actors?" The essence behind calling a ban on these weapons now becomes the scapegoat for any miscalculation that the machine might do.

International Committee of the Red Cross (ICRC) details out some views on humanitarian laws and robots as, "No matter how sophisticated weapons become, it is essential that these are in line with the rules

of war. International humanitarian law is all about making choices that preserve a minimum of human dignity in times of war and make sure that living together again is possible once the last bullet has been shot."

Even the United Nations meeting in Geneva earlier this year saw delegates from over 90 countries discussing the future of autonomous weapons. *IEEE Spectrum* states that experts believe autonomous weapons are potentially weapons of mass destruction. The United States' Directive 3000.09 prevents the use of fully-autonomous weapons but allows their development.

## What could be done

"Weapons have been banned before, but for autonomous weapons the control is important," says Dr Peter Asaro, co-founder and vice chair, International Committee for Robot Arms Control. On the argument of stopping progress in the field, he explains, "Banning biological and chemical weapons did not stop progress in the fields. We just stopped weaponising these."

Some argue that these systems would be unable to adhere to the current laws of war. War is brutal, and trying to create guidelines in war does not work. Dr Asaro believes, "It is great to think that technology can do better but, at the moment, we do not have such efficient systems." So should the priority not be to stop war and all weapons instead of making a fuss about autonomous weapons?

Naik recalls a very interesting quote by George Bernard Shaw, "A native American elder once described his own inner struggles in this manner: Inside of me there are two dogs. One of the dogs is mean and evil. The other dog is good. The mean dog fights the good dog all the time. When asked which dog wins, he reflected for a moment and replied: The one I feed the most." **EFY**